Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 26

Existence Proofs

Covers Chartrand & Zhang, Section 3.6

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Two jobs, very different costs

Prove $\exists x \in S,\ R(x)$**:** name one element and verify it.
Disprove $\exists x \in S,\ R(x)$**:** argue that $R(x)$ fails for *every* $x \in S$.
The first is often a one-liner. The second is a genuine universal argument — testing values does not do it.

Three mistakes to expect

1. **Naming a witness without verifying it.**
2. **"Disproving" an existence claim by testing a few values.**
3. **Choosing a witness too late.** In $\exists r\ \forall s$, the r must work for every s at once.

Part II: Practice Problems

Section A — Constructive existence

1. Prove each by exhibiting a witness and verifying it.

(a) There exists an integer n such that $50 - n^2 < 0$.

(b) There exists a real number x such that $x^2 - 3x - 10 = 0$.

(c) There exist integers m and n of opposite parity with $(m-3)^2 + (n-5)^2 \leq 1$.

Section B — Disproving existence

2. Disprove each.

(a) There exists a real number x such that $x^2 + 6x + 10 = 0$.

(b) There exists an integer n such that $n^2 = 3$.

Section C — Mixed quantifiers

3. Prove or disprove each.

(a) For every integer n, there exists an integer m with $n + m = 0$.

(b) There exists an integer m such that for every integer n, $n + m = n$.

(c) There exists an integer m such that for every integer n, $n + m = 0$.

Section D — Non-constructive existence

4. Prove: there exist irrational numbers a and b such that $a + b$ is rational.

5. Prove that among any 13 integers, some two leave the same remainder on division by 12.

Section E — Writing

6. In five to eight complete sentences, explain the difference between a constructive and a non-constructive existence proof. Use Result 3.38 as your example of the second, and say why the distinction matters to a computer scientist.